

Housing price prediction

Dataset: kaggle Housing price 545 rows, 13 columns

1. Dataset & Cleaning

The dataset contains 545 rows and 13 columns describing house features. Used to predict price. No missing value or duplicated rows were found. Six binary Yes/No col were encoded to 1/0, and furnishing status was one-hot encoded.

2. Key visual findings

- * Price distribution is right-skewed - most homes are moderately priced, with a few high-value outlier pulling the average up.
- * Correlation heatmap shows area, bathrooms, and air conditioning have the strongest positive relationship with price.
- * Actual vs predicted scatter plot shows Random forest prediction clustering tightly around the ideal fit line, confirm model accuracy.

3. Model Performance

Metric	Linear Regression	Random forest
MSE	970,000	700,000
RMSE	1,300,000	970,000
R^2 score	0.65	0.80

4. Insights & Summary

- Area, number of bathrooms and air conditioning were the strongest driver.
- Rf explained about 80% of variation in house prices ($R^2 = 0.80$)
- Bedroom count barely affected price, even though it's usually the first thing buyers look at. Comfort feature like AC mattered more.